

MUSHFIQUE TANZIM MUZTABA

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate at The Hong Kong Polytechnic University with experience in mechanical design (SOLIDWORKS, Finite Element Analysis), robotics (ROS2, SLAM, Nav2), AI/computer vision (YOLO, OpenCV), and engineering workflow automation (Power Platform). Seeking graduate opportunities in mechanical engineering, automation, and product/system engineering.

EDUCATION

The Hong Kong Polytechnic University

Hong Kong

Bachelor of Engineering (Honours) in Mechanical Engineering

Graduation Date: May 2026

- **Accolades:** The Hong Kong Polytechnic University Entry Scholarship (Academic)
- **Relevant Coursework:** Fluid Mechanics, Engineering Thermodynamics, Numerical Methods for Engineers, Linear Systems and Control, Manufacturing Fundamentals, Fundamentals of Robotics

WORK EXPERIENCE

CLP Power Hong Kong

Hong Kong

Sandwich Intern

July 2024 – June 2025

Project Execution Team (PET)

- Built a Power Apps 'Lessons Learned' portal for 100+ members across 4 business groups to centralize and share project insights, with planned wider rollout.
- Created engagement features (likes, leaderboards) and multi-stage approval workflows in Power Automate, cutting feedback turnaround time.
- Supported project governance activities and completed CLP Project Management Academy training.

Hong Kong Battery Energy Storage System (HKBESS)

- Developed a Power Apps Management-of-Change tool for the Hung Shui Kiu "A" substation, centralizing engineering change requests.
- Engineered multi-level approval routing in Power Automate, reducing cycle time by 30% with real-time Teams notifications.
- Documented 15+ operational risks for the battery storage system, informing project safety protocols.

MangDang Technology Co., Limited

Hong Kong

Contract Robotics Intern (Part-time)

February 2026 – March 2026

- Migrated quadruped robot software stack from ROS2 Humble to Jazzy, including simulator transition from Gazebo Classic to Gazebo Harmonic and full Ubuntu 24.04 compatibility.
- Re-engineered the CHAMP quadruped contact sensor node from Gazebo transport APIs to GZ Harmonic, improving gait control accuracy.
- Validated autonomous navigation by testing SLAM (Cartographer, SLAM Toolbox), Nav2, and teleoperation on both simulator and Raspberry Pi hardware.

Thai German Graduate School of Engineering (TGGs)

Bangkok, Thailand

Research Intern

August 2023

- Designed a motorbike crash-test dolly in SOLIDWORKS from concept sketches to production-ready assemblies.
- Performed iterative FEA under simulated crash loads, optimizing geometry for target safety factors and material efficiency.
- Engineered an energy-absorbing crash-cushion system, reducing peak impact forces and improving test repeatability.

PROJECT EXPERIENCE

Autonomous Indoor Thermal Inspection Robot

January 2026 – May 2026

- Developed an autonomous indoor inspection robot using ROS 2, SLAM Toolbox, and frontier-based exploration with LiDAR, RGB, and thermal sensors for non-contact anomaly detection (>35°C).
- Built a multi-layer perception pipeline combining thermal hotspot detection, object detection (YOLO11s), and Gemini 2.5 Flash multimodal analysis on a Raspberry Pi 5 with distributed processing via rosbridge.
- Implemented a natural-language control interface (Gemini 2.5 Flash) and operator dashboard (React) with live SLAM map, camera feeds, and thermal visualization.

Autonomous Vision-Guided Targeting System

November 2025

- Developed an autonomous targeting system in Python/OpenCV on Raspberry Pi, achieving 90% detection accuracy under variable lighting via HSV colour thresholds.
- Implemented PID controller ($K_p=0.12$, $K_i=0.005$, $K_d=0.25$) with frame smoothing and dead zones for precise 2-axis servo control via I2C PWM.
- Integrated trajectory compensation and GPIO-controlled firing logic for reliable target acquisition at 1 m.

AI Multi-Model Object Detection System

October 2025

- Trained a YOLOv11n model to 90–94% accuracy on a custom 100+ image dataset (PyTorch, Roboflow).
- Combined YOLO detection with HSV colour classification for multi-attribute identification (82–99% confidence across categories).
- Built a complete ML pipeline from data labelling to model deployment using Python and OpenCV.

Personal Portfolio Website

August 2025

- Developed a responsive engineering portfolio using React 19, TypeScript, and Vite to showcase robotics and mechanical design projects.
- Engineered interactive features including scroll-based animations (Framer Motion) and a custom Canvas particle background, with smooth scrolling and full mobile responsiveness.

PolyU E Formula Racing Team

September 2023 – May 2024

- Optimised suspension geometry using Lotus Suspension Analysis to evaluate handling characteristics for the team's electric formula vehicle.
- Designed tie rods, steering column, and steering linkages in SOLIDWORKS, ensuring manufacturability and assembly fit.

Automated Material-Handling Trolley

September 2023

- Designed and manufactured an automated industrial trolley in SOLIDWORKS with motorized drive and steering mechanisms.
- Implemented autonomous loading/unloading, straight-line tracking, and 90° turning within a fixed-footprint environment.

TECHNICAL SKILLS

- **CAD and Mechanical Design:** SOLIDWORKS (modelling, FEA, assemblies), AutoCAD
- **Programming:** Python, C++, MATLAB
- **Engineering Tools:** Microsoft Power Apps, Power Automate, Git, Office 365
- **Robotics:** ROS2 (Humble, Jazzy), Gazebo, SLAM Toolbox, Cartographer, Nav2
- **AI and Computer Vision:** PyTorch, YOLOv11, OpenCV
- **Embedded Systems:** Raspberry Pi, STM32, I2C, SPI, GPIO
- **Web Development:** React, TypeScript, Tailwind CSS, Vite, Framer Motion

LICENSES AND CERTIFICATIONS

- Hong Kong Green Card (Construction Industry Safety Training) *Valid until March 2028*
- Hong Kong Blue Card (Shipboard Cargo Handling Basic Safety Training) *Valid until January 2028*

LinkedIn Learning Certificates: SOLIDWORKS: Advanced Engineering Drawings, SOLIDWORKS: Advanced Simulation, SOLIDWORKS: Modelling Gears, Learning AutoCAD 2024.

LEADERSHIP AND ACTIVITIES

PolyU ENGL English Debate Club

Hong Kong

Creative Designer

August 2024 – June 2025

- Designed posters, social media graphics, and event merchandise, strengthening brand engagement for 500+ members.

PolyU International Student Association

Hong Kong

Creative Designer and Social Media Manager

July 2023 – June 2024

- Created promotional graphics and videos garnering 15,000+ views, boosting event participation.

PolyU Student Halls of Residence

Hong Kong

Event Organizer

November 2022 – June 2025

- Organized and coordinated inter-hall activities for 120+ participants, fostering community engagement.